Brief

WRITE UP

Data Quality Issues Identified & Fixes Applied

Overview

Before building the Power BI dashboard, I performed a detailed data cleaning and preprocessing process on the raw Marketing and Shopify datasets to ensure data accuracy, consistency, and reliable reporting.

1. Missing Values (Null / Blank / NaN)

Issue Found:

The dataset contained a high number of blank, null, and NaN values across text, numerical, and count-based columns.

Fix Applied:

Text Columns (Country Funnel, Geo Location Segment, Billing Country, Platform, etc.):
Replaced missing values with "Unknown" to preserve category structure without deleting rows.

Numerical Columns (Gross Sales, Net Sales, Total Sales, Spend, Revenue, etc.):
Replaced null values using the Median to reduce outlier distortion.

Count Columns (Clicks, Impressions, Orders, Likes, etc.):
Replaced null values with 0 because missing count values logically indicate no recorded activity.

2. Negative Values in Count-Based Columns

Issue Found:

Several count-related columns such as Clicks, Impressions, Orders, and Likes contained negative values, which are logically invalid.

Fix Applied:

Replaced all negative values in count-based columns with 0

Used conditional checks to identify and standardize invalid records

3. Duplicate Records

<u>Issue Found:</u>

The raw dataset contained duplicate rows that could inflate KPIs such as sales, conversions, and campaign performance.

<u>Fix Applied:</u>

Removed duplicate rows using Power Query's Remove Duplicates function

Ensured data uniqueness based on relevant identifiers

4. Inconsistent Date Formats

<u>Issue Found:</u>

Date columns had inconsistent formatting and missing values, which could impact trend analysis.

<u>Fix Applied:</u>

Standardized all date columns into a consistent date format

Validated campaign date logic:

Campaign Start Date: 01-Jan-2026

Campaign End Date: 28-Mar-2026

Ensured start dates occurred before end dates

5. High Null Percentage in Key Columns

Issue Found:

Order ID and Product ID columns contained nearly 50% null values.

Fix Applied:

Preserved rows where business context was still useful

Replaced blanks where appropriate

Avoided unnecessary deletion to prevent excessive data loss

6. Text Standardization

Issue Found:

Categorical columns had inconsistent text formatting, blanks, and mixed structures.

Fix Applied:

Standardized text values

Replaced blanks with "Unknown"

Improved segmentation consistency for dashboard slicing

7. Data Modeling Challenge (Many-to-Many Relationship)

<u>Issue Found:</u>

Direct relationship between Campaign and Shopify tables created many-to-many relationship issues.

<u>Fix Applied:</u>

Created a dynamic Calendar Table using minimum campaign start date and maximum campaign end date

Used Calendar Table as a bridge between datasets

Established proper one-to-many relationships for accurate time intelligence

8. KPI Recalculation & Validation

Issue Found:

Raw performance metrics required validation.

Fix Applied:

Recalculated key metrics using DAX:

CTR = Clicks / Impressions

CPC = Spend / Clicks

CPM = (Spend / Impressions) × 1000

ROI = (Revenue - Spend) / Spend

ROAS = Revenue / Spend

This ensured business metrics were accurate and standardized.

Final Outcome

After cleaning, transforming, and validating the dataset:

Improved data consistency

Reduced reporting errors

Built a reliable data model

Enabled accurate dashboard insights across campaign, channel, regional, and audience analysis

Conclusion

This cleaning process transformed raw, inconsistent data into a structured, analysis-ready dataset, ensuring that all dashboard insights were based on reliable and business-valid information.

THANK YOU